

Run 04 | TEA

India

IV only

Target Total early-stage entrepreneurial activity; binary 1 vs. 0
Sample 27,555 respondents **Coverage** 1 country; 2015–2022
Positive cases 3,188 (11.57%) **Case time** 32s
Train / val. / test 18,739 / 3,306 / 5,510 (68% / 12% / 20%, stratified)
Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.
Fraser EFW predictors (7): Overall economic freedom, Economic freedom quartile, Size of government, Legal system and property rights, Sound money, Freedom to trade internationally, Regulation. Total independent variables: 20.
Controls (0): None. Country and respondent ID are excluded as predictors.
Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: CatBoost (0.9973). Highest F1: Decision Tree (0.9347). The all-negative accuracy baseline is 88.43% (full sample).

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
CatBoost	0.9973	0.9767	98.33	91.35	94.51	0.9290	5s
XGBoost	0.9972	0.9753	98.40	90.67	96.08	0.9329	<1s
Random Forest	0.9971	0.9741	98.38	91.90	94.35	0.9311	1s
H2O GBM	0.9971	0.9731	98.44	90.69	96.39	0.9346	3s
Logistic Regression	0.9969	0.9720	98.38	91.39	94.98	0.9315	<1s
Elastic Net	0.9969	0.9716	98.38	91.64	94.66	0.9313	7s
Lasso	0.9969	0.9715	98.38	91.64	94.66	0.9313	5s
Ridge	0.9966	0.9688	98.37	90.16	96.39	0.9317	7s
Decision Tree	0.9941	0.9209	98.46	91.57	95.45	0.9347	<1s
Naive Bayes	0.9344	0.6725	93.36	67.76	81.16	0.7386	3s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	1,823.11	59.23
2	Established business owner	637.85	20.72
3	Knowing entrepreneurs	67.55	2.19
4	Business angel	63.58	2.07
5	Business exit	46.69	1.52
6	Self-efficacy	40.07	1.30
7	Business exit (missing indicator)	33.42	1.09
8	Freedom to trade internationally	29.34	0.95

Share is the percentage of total forest impurity importance across all 38 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Fraser EFW predictors within the same forest

Rank	Fraser EFW predictor	Raw importance	Share%
1	Freedom to trade internationally	29.34	0.95
2	Overall economic freedom	24.51	0.80
3	Legal system and property rights	21.34	0.69

Fraser EFW subset ranked by raw importance; shares use total forest importance, not the Fraser EFW subtotal.

Source: Completed Objective 3 batch, 29 September 2026; 04_obj3_india_iv_TEA. Scores and importance are taken directly from saved CSV outputs.